Notes on Gauge Theory

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Preface

本次讨论班的内容主要是规范理论，我们将会介绍规范理论所需的几何基础、用几何语言描述规范理论，以及用规范理论研究低维拓扑。

首先我们会介绍规范理论的几何基础，如联络、主丛、特征类等，这些内容在物理学中是非常重要的工具，在数学中也有着广泛的应用。我们将会从黎曼几何中的 Levi-Civita 联络开始，逐渐引入更一般的仿射联络和主丛上的联络，并介绍它们的曲率和特征类等相关内容。

物理学家用联络描述规范场论中的规范场，曲率则对应场强，我们将会用几何语言描述电磁理论和 Yang-Mills 理论等规范场论的基本内容。

最后，数学家用规范理论构造了很多重要的拓扑不变量，如 Donaldson 不变量、Seiberg-Witten 不变量等，我们将会介绍这些不变量的定义，以及它们在低维拓扑中的应用。

以下是本次讨论班的参考资料：

- 《Differential Geometry - Connections, Curvature and Characteristic Classes》(Loring W. Tu)
- 《黎曼几何引论 (上下两册)》(陈维桓，李兴校)
- 《Lectures on Seiberg-Witten Invariants》(John Douglas Moore)
- 何思奇老师在陈所暑期学校的讲义 http://www.cim.nankai.edu.cn/2020/0709/c11453a284191/page.htm

1 Prologue: Electromagnetism and $U(1)$ Gauge Theory

1.1 Maxwell's equations

We begin the story with Maxwell's equations. Recall that Maxwell's equations in vacuum are

$$\begin{cases} \operatorname{div} \mathbf{E} = 0, \\ \operatorname{div} \mathbf{B} = 0, \\ \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{curl} \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases}$$

Here

$$\begin{aligned} \mathbf{E} &= (E_x, E_y, E_z) \quad \text{is the electric field,} \\ \mathbf{B} &= (B_x, B_y, B_z) \quad \text{is the magnetic field.} \end{aligned}$$

Each component is a function of the spacetime coordinates (t, x, y, z).

Below we will list some concepts and notations commonly used in physics:

- 4-*vector potential* A_μ[1]

  $$A_\mu = (\phi, A_x, A_y, A_z),$$

  where ϕ is the *scalar potential* and $\mathbf{A} = (A_x, A_y, A_z)$ is the *vector potential.*

- *Electromagnetic field strength tensor* $F_{\mu\nu}$

  $$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix}.$$

- Relation between A_μ and $F_{\mu\nu}$

  $$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu{}^{2} \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

- *Gauge transformation of the* 4-*vector potential*

  $$A_\mu \mapsto A_\mu + \partial_\mu \chi \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \operatorname{grad} \chi. \end{cases}$$

  where χ is an arbitrary smooth function. One can check that this transformation leaves the electromagnetic field strength tensor $F_{\mu\nu}$ invariant.

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[1] $\mu = 0, 1, 2, 3$ are the spacetime indices, where 0 corresponds to the time coordinate t, and $1, 2, 3$ correspond to the spatial coordinates x, y, z.

[2] Here we use the notation $\partial_0 := \frac{\partial}{\partial t}, \partial_1 := \frac{\partial}{\partial x}, \partial_2 := \frac{\partial}{\partial y}, \partial_3 := \frac{\partial}{\partial z}$.

Although Maxwell's equations are already quite concise, we can still transform them into an even more elegant expression using the language of differential forms.

Let $\mathbb{R}^{3,1}$ be the spacetime with coordinates (t, x, y, z), equipped with the Minkowski metric

$$\mathrm{d}s^2 = \mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2.$$

For simplicity, we set the speed of light $c = 1$.

The Hodge star operator $*$ on $\Omega^*(\mathbb{R}^{3,1})$ is defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z, \quad \forall \alpha, \beta \in \Omega^*(\mathbb{R}^{3,1}).$$

Explicitly, we have

$$\begin{aligned}
*1 &= \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,\\
*\mathrm{d}t &= \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,\\
*\mathrm{d}x &= \mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z,\\
*\mathrm{d}y &= \mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x,\\
*\mathrm{d}z &= \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y,\\
*(\mathrm{d}t \wedge \mathrm{d}x) &= -\mathrm{d}y \wedge \mathrm{d}z,\\
*(\mathrm{d}t \wedge \mathrm{d}y) &= -\mathrm{d}z \wedge \mathrm{d}x,\\
*(\mathrm{d}t \wedge \mathrm{d}z) &= -\mathrm{d}x \wedge \mathrm{d}y,\\
*(\mathrm{d}x \wedge \mathrm{d}y) &= \mathrm{d}t \wedge \mathrm{d}z,\\
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*(\mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y) &= \mathrm{d}z,\\
*(\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z) &= \mathrm{d}x,\\
*(\mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x) &= \mathrm{d}y,\\
*(\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z) &= \mathrm{d}t,\\
*(\mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z) &= -1.
\end{aligned}$$

It is direct to check that

$$*^2 = (-1)^{p+1} \quad \text{on } \Omega^p(\mathbb{R}^{3,1}).$$

Now we can introduce the electromagnetic field strength 2-form

$$F = -\mathrm{d}t \wedge (E_x \mathrm{d}x + E_y \mathrm{d}y + E_z \mathrm{d}z) + B_x \mathrm{d}y \wedge \mathrm{d}z + B_y \mathrm{d}z \wedge \mathrm{d}x + B_z \mathrm{d}x \wedge \mathrm{d}y.$$

Then

$$\begin{aligned}
\mathrm{d}F =& \mathrm{d}t \wedge \Big[(\partial_x E_y - \partial_y E_x)\mathrm{d}x \wedge \mathrm{d}y + (\partial_y E_z - \partial_z E_y)\mathrm{d}y \wedge \mathrm{d}z + (\partial_z E_x - \partial_x E_z)\mathrm{d}z \wedge \mathrm{d}x\Big]\\
&+ \mathrm{d}t \wedge \Big(\partial_t B_x \mathrm{d}y \wedge \mathrm{d}z + \partial_t B_y \mathrm{d}z \wedge \mathrm{d}x + \partial_t B_z \mathrm{d}x \wedge \mathrm{d}y\Big)\\
&+ \Big(\partial_x B_x + \partial_y B_y + \partial_z B_z\Big)\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z,
\end{aligned}$$

Thus

$$\begin{cases} \text{div } \mathbf{B} = 0, \\ \text{curl } \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \end{cases} \iff \mathrm{d}F = 0.$$

On the other hand,

$$*F = E_x \mathrm{d}y \wedge \mathrm{d}z + E_y \mathrm{d}z \wedge \mathrm{d}x + E_z \mathrm{d}x \wedge \mathrm{d}y + \mathrm{d}t \wedge (B_x \mathrm{d}x + B_y \mathrm{d}y + B_z \mathrm{d}z).$$

So

$$\begin{aligned} \mathrm{d} * F = & - \mathrm{d}t \wedge \Big[(\partial_x B_y - \partial_y B_x)\mathrm{d}x \wedge \mathrm{d}y + (\partial_z B_x - \partial_x B_z)\mathrm{d}z \wedge \mathrm{d}x + (\partial_y B_z - \partial_z B_y)\mathrm{d}y \wedge \mathrm{d}z\Big] \\ & + \mathrm{d}t \wedge \Big(\partial_t E_x \mathrm{d}y \wedge \mathrm{d}z + \partial_t E_y \mathrm{d}z \wedge \mathrm{d}x + \partial_t E_z \mathrm{d}x \wedge \mathrm{d}y\Big) \\ & + \Big(\partial_x E_x + \partial_y E_y + \partial_z E_z\Big)\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z. \end{aligned}$$

Thus

$$\begin{cases} \text{div } \mathbf{E} = 0, \\ \text{curl } \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \mathrm{d} * F = 0.$$

Since $* : \Omega^p(\mathbb{R}^{3,1}) \to \Omega^{4-p}(\mathbb{R}^{3,1})$ is an isomorphism,

$$\mathrm{d} * F = 0 \iff *\mathrm{d} * F = 0.$$

Therefore,

$$\begin{cases} \text{div } \mathbf{E} = 0, \\ \text{div } \mathbf{B} = 0, \\ \text{curl } \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{curl } \mathbf{B} = \frac{\partial \mathbf{E}}{\partial t}. \end{cases} \iff \begin{cases} \mathrm{d}F = 0, \\ \mathrm{d}^* F = 0. \end{cases}$$

Here $\mathrm{d}^* = *\mathrm{d}*$ is the dual of d.

In Riemannian geometry, a p-form α is called *harmonic* if $\mathrm{d}\alpha = 0$ and $\mathrm{d}^*\alpha = 0$. Thus Maxwell's equations in vacuum corresponds to a harmonic 2-form F.

Poincaré lemma says all closed forms on $\mathbb{R}^{3,1}$ are exact. So if $\mathrm{d}F = 0$, there exists a 1-form A such that

$$F = \mathrm{d}A.$$

If we write

$$A = \phi \mathrm{d}t + A_x \mathrm{d}x + A_y \mathrm{d}y + A_z \mathrm{d}z,$$

Then

$$F = \mathrm{d}A \implies \begin{cases} E_x = \partial_x \phi - \partial_t A_x, \\ E_y = \partial_y \phi - \partial_t A_y, \\ E_z = \partial_z \phi - \partial_t A_z, \\ B_x = \partial_y A_z - \partial_z A_y, \\ B_y = \partial_z A_x - \partial_x A_z, \\ B_z = \partial_x A_y - \partial_y A_x. \end{cases} \implies \begin{cases} \mathbf{E} = \operatorname{grad} \phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} = \operatorname{curl} \mathbf{A}. \end{cases}$$

It is easy to see that A is not unique. In fact, if we transform $A \mapsto A + \mathrm{d}\chi$ for some smooth function χ, then

$$F \mapsto \mathrm{d}A + \mathrm{d}^2\chi = \mathrm{d}A = F.$$

Explicitly,

$$A \mapsto A + \mathrm{d}\chi \implies \begin{cases} \phi \mapsto \phi + \partial_t \chi, \\ A_x \mapsto A_x + \partial_x \chi, \\ A_y \mapsto A_y + \partial_y \chi, \\ A_z \mapsto A_z + \partial_z \chi. \end{cases} \implies \begin{cases} \phi \mapsto \phi + \frac{\partial \chi}{\partial t}, \\ \mathbf{A} \mapsto \mathbf{A} + \nabla \chi. \end{cases}$$

We can summarize the above discussion in a table:

Physics	**Differential Geometry**
4-vector potential A_μ	1-form A
Electromagnetic field strength tensor $F_{\mu\nu}$	2-form F
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	$F = \mathrm{d}A$
Gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \chi$	$A \mapsto A + \mathrm{d}\chi$
Maxwell's equations	$\mathrm{d}F = 0, \mathrm{d}^* F = 0$

1.2 $U(1)$ gauge theory

Let $\psi(x)$ be a complex-valued function on $\mathbb{R}^{3,1}$, which represents the wave function of a matter field. Here x denotes the spacetime coordinates. In physics, $|\psi(x)|^2$ represents the probability amplitude of finding the particle at a given point in spacetime. If we transform

$$\psi(x) \mapsto \psi'(x) = e^{i\chi}\psi(x)$$

for some real constant χ. This transformation is called a *global $U(1)$ gauge transformation*. Since the probability amplitude $|\psi'(x)|^2 = |\psi(x)|^2$ remains the same, the physics is invariant under global $U(1)$ gauge transformations.

Now we want to promote the global $U(1)$ gauge symmetry to a local one. That is, we want the physics to be invariant under the transformation

$$\psi(x) \mapsto \psi'(x) = e^{i\chi(x)}\psi(x)$$

for some smooth function $\chi \in C^\infty(\mathbb{R}^{3,1})$.

For some physical reason, we want the derivative of ψ to be covariant under transformations. This holds in the case of global $U(1)$ gauge transformations, since

$$\mathrm{d}\psi \mapsto \mathrm{d}\psi' = e^{i\chi}\mathrm{d}\psi.$$

However, in the case of local $U(1)$ gauge transformations,

$$\mathrm{d}\psi \mapsto \mathrm{d}\psi' = e^{i\chi(x)}\mathrm{d}\psi + (\mathrm{d}e^{i\chi(x)})\psi.$$

The extra term $(\mathrm{d}e^{i\chi(x)})\psi$ breaks the covariance of the derivative.

To fix this problem, we introduce a 1-form A and define the *covariant derivative* by

$$\nabla\psi = \mathrm{d}\psi + iA\psi.$$

Under the local $U(1)$ gauge transformation,

$$\nabla \mapsto \nabla' = \mathrm{d} + iA',$$

we want $\nabla\psi$ to be covariant, i.e.

$$\nabla\psi \mapsto \nabla'\psi' = e^{i\chi(x)}\nabla\psi.$$

Then by

$$\begin{aligned} \nabla'\psi' = \mathrm{d}\psi' + iA'\psi' &= e^{i\chi(x)}\mathrm{d}\psi + ie^{i\chi(x)}(A' + \mathrm{d}\chi)\psi, \\ e^{i\chi(x)}\nabla\psi = e^{i\chi(x)}(\mathrm{d}\psi + iA\psi) &= e^{i\chi(x)}\mathrm{d}\psi + ie^{i\chi(x)}A\psi, \end{aligned}$$

we must have

$$A' = A - \mathrm{d}\chi.$$

Ignoring the sign, this is exactly the gauge transformation of the electromagnetic potential A in electromagnetism. Thus, the electromagnetic potential A can be interpreted as a gauge field that ensures the covariance of the derivative under local $U(1)$ gauge transformations.

2 Affine Connections

2.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

2.1.1 Covariant Derivative

Definition 2.1. A **connection** ∇ on a mannifold M is a map

$$\begin{aligned}\nabla : C^\infty(M,TM) \times C^\infty(M,TM) &\to C^\infty(M,TM)\\ (Y,X) &\mapsto \nabla_Y X\end{aligned}$$

satisfying the following properties:

(C1) $$\nabla_{f_1Y_1+f_2Y_2}X = f_1\nabla_{Y_1}X + f_2\nabla_{Y_2}X,$$

(C2) $$\nabla_Y(\lambda_1X_1+\lambda_2X_2) = \lambda_1\nabla_YX_1 + \lambda_2\nabla_YX_2,$$

(C3) $$\nabla_Y fX = \mathrm{d}f(Y)\cdot X + f\nabla_YX.$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 2.2. The **torsion** of a connection ∇ is defined as the tensor

$$T(X,Y) := \nabla_XY - \nabla_YX - [X,Y].$$

The **curvature** of a connection ∇ is defined as the tensor

$$R(X,Y)Z := \nabla_X\nabla_YZ - \nabla_Y\nabla_XZ - \nabla_{[X,Y]}Z.$$

Remark. The torsion T can only be defined for connections on the tangent bundle, since it involves both ∇_XY and ∇_YX.

Definition 2.3. A connection ∇ of a Riemannian manifold (M,g) is called a **Levi-Civita connection** if it is torsion-free and compatible with the metric g, i.e.

(LC1) $$\nabla_XY - \nabla_YX = [X,Y],$$

(LC2) $$X(g(Y,Z)) = g(\nabla_XY,Z) + g(Y,\nabla_XZ).$$

Exterior convariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned}\nabla : C^\infty(M,TM) &\to C^\infty(M,T^*M\otimes TM)\\ X &\mapsto \nabla X\end{aligned}$$

where ∇X is a 1-form taken value in TM[1] and $(\nabla X)(Y) := \nabla_YX$. ∇ is not a $C^\infty(M)$-linear map, instead,

$$\nabla(fX) = \mathrm{d}f\otimes X + f\nabla X.$$

[1]One can simply view ∇X as a $(1,1)$-tensor field.

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M,TM) &\to \Omega^{*+1}(M,TM)^1\\ \eta &\mapsto \nabla\eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha\wedge\eta) = \mathrm{d}\alpha\wedge\eta + (-1)^q\alpha\wedge\nabla\eta, \quad \forall\alpha\in\Omega^q(M), \eta\in\Omega^*(M,TM),$$

For example, let $\eta = \alpha\otimes X$ be a section of $\Omega^r(M,TM)$, then

$$\nabla\eta = \nabla(\alpha\otimes X) = \mathrm{d}\alpha\otimes X + (-1)^r\alpha\wedge\nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla\circ\nabla, \nabla^3, \dots, \nabla^k$, and we can prove the following property:

Proposition 2.4. *$\nabla^2 : \Omega^*(M,TM) \to \Omega^{*+2}(M,TM)$ is an $\Omega^*(M)$-linear map, i.e. for any $\alpha\in\Omega^*(M), \eta\in\Omega^*(M,TM)$,*

$$\nabla^2(\alpha\wedge\eta) = \alpha\wedge\nabla^2\eta.$$

*Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \mathrm{End}(TM))$, which is called the **curvature** of the connection ∇.*

Proof.

$$\begin{aligned}\nabla^2\big(\alpha\wedge\eta\big) =& \nabla\big(\mathrm{d}\alpha\wedge\eta + (-1)^q\alpha\wedge\mathrm{d}^\nabla\eta\big)\\ =& \mathrm{d}^2\alpha\wedge\eta + (-1)^{q+1}\mathrm{d}\alpha\wedge\mathrm{d}^\nabla\eta + (-1)^q\mathrm{d}\alpha\wedge\nabla\eta + \alpha\wedge\nabla^2\eta\\ =& \alpha\wedge\nabla^2\eta.\end{aligned}$$

□

Unlike the usual exterior derivative d, we have $\nabla^2 \neq 0$ in general. The following proposition explains why ∇^2 is called the curvature of ∇:

Proposition 2.5. *For $X, Y, Z\in C^\infty(M,TM)$, we have*

$$(\nabla^2 Z)(X,Y) = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

Proof. We will prove it after introducing the connection 1-forms. □

2.1.2 Connection 1-form

Locally, we always have a frame. Let $\{e_i\}$ be a local frame of TM on an open set $U\subset M$, and $\{\theta^i\}$ be the dual frame. then we can write

$$\nabla_{e_i}e_j = \Gamma^k_{ij}e_k,$$

where Γ^k_{ij} are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega^k_j := \Gamma^k_{ij}\theta^i,$$

[1] $\Omega^r(M,TM) := C^\infty(M, \wedge^r T^*M\otimes TM)$.

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)$[1], then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_n) = (e_1, \dots, e_n)\,\omega = (e_1, \dots, e_n) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \cdots & \omega_n^n \end{pmatrix}$$ [2]

For any vector field $X = X^i e_i$, we have

$$\begin{aligned} \nabla X =& \nabla\left((e_1, \dots, e_n) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right) \\ =& \Big(\nabla(e_1, \dots, e_n)\Big) \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + (e_1, \dots, e_n) \begin{pmatrix} \mathrm{d}X^1 \\ \vdots \\ \mathrm{d}X^n \end{pmatrix} \\ =& (e_1, \dots, e_n) \left(\mathrm{d} \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} + \omega \begin{pmatrix} X^1 \\ \vdots \\ X^n \end{pmatrix} \right). \end{aligned}$$

Thus under a local frame, ∇ can be expressed as

$$\nabla = \mathrm{d} + \omega.$$

We can compute the curvature in terms of the connection 1-form ω:

$$\begin{aligned} \nabla^2(e_1, \dots, e_n) =& \nabla\big(\nabla(e_1, \dots, e_n)\big) = \nabla\big((e_1, \dots, e_n)\,\omega\big) \\ =& \big(\nabla(e_1, \dots, e_n)\big) \wedge \omega + (e_1, \dots, e_n)\,\mathrm{d}\omega \\ =& (e_1, \dots, e_n)\,\omega \wedge \omega + (e_1, \dots, e_n)\,\mathrm{d}\omega \\ =& (e_1, \dots, e_n)\,\big(\mathrm{d}\omega + \omega \wedge \omega\big). \end{aligned}$$

Define

$$\Omega := \mathrm{d}\omega + \omega \wedge \omega, \tag{2.1}$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 2.6. *Let $R(e_i, e_j)e_k = R_{ijk}^l e_l$, then*

$$\Omega_k^l(e_i, e_j) = R_{ijk}^l.$$

[1] The upper index k is the row index, and the lower index j is the column index.

[2] This is an awkward notation. Since we arrange the frame as row vectors so matirx ω should multiply on the right. For a vector field X and a differential forms $\alpha \in \Omega^q(M)$, we have $X \otimes \alpha = \alpha \otimes X$. For a general $\eta \in \Omega^p(M, TM)$, we have $\eta \otimes \alpha = (-1)^{pq} \alpha \otimes \eta$.

Proof. Since $\Omega^l_k = \mathrm{d}\omega^l_k + \omega^l_m \wedge \omega^m_k$,

$$\begin{aligned}(\mathrm{d}\omega^l_k)(e_i, e_j)e_l &= e_i(\omega^l_k(e_j))e_l - e_j(\omega^l_k(e_i))e_l - \omega^l_k([e_i, e_j])e_l \\ =&\nabla_{e_i}(\omega^l_k(e_j)e_l) - \omega^l_k(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega^l_k(e_i)e_l) + \omega^l_k(e_i)\nabla_{e_j}e_l - \omega^l_k([e_i, e_j])e_l \\ =&\nabla_{e_i}\nabla_{e_j}e_k - \omega^l_k(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega^l_k(e_i)\nabla_{e_j}e_l - \omega^l_k([e_i, e_j])e_l,\end{aligned}$$

and

$$\begin{aligned}(\omega^l_m \wedge \omega^m_k)(e_i, e_j)e_l =&\omega^l_m(e_i)\omega^m_k(e_j)e_l - \omega^l_m(e_j)\omega^m_k(e_i)e_l \\ =&\omega^m_k(e_j)\nabla_{e_i}e_m - \omega^m_k(e_i)\nabla_{e_j}e_m.\end{aligned}$$

Thus

$$\Omega^l_k(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i,e_j]}e_k = R(e_i, e_j)e_k. \qquad \square$$

Corollary 2.7. *This proposition immediately implies Prop 2.5 since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega^l_k(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

Let $\{\tilde{e_j}\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e_j}\}$ is $a = (a^j_i)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n)\, a,$$

then

$$\begin{aligned}(\tilde{e}_1, \dots, \tilde{e}_n)\tilde{\omega} =&\nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla\big((e_1, \dots, e_n)\, a\big) \\ =&\big(\nabla(e_1, \dots, e_n)\big)a + (e_1, \dots, e_n)\mathrm{d}a \\ =&(e_1, \dots, e_n)\omega a + (e_1, \dots, e_n)\mathrm{d}a \\ =&(\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + \mathrm{d}a).\end{aligned}$$

Thus

$$\tilde{\omega} = a^{-1}\mathrm{d}a + a^{-1}\omega a. \tag{2.2}$$

Applying the transformation formula (2.2) to (2.1), we get

$$\begin{aligned}\tilde{\Omega} =&\mathrm{d}\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega} \\ =&\mathrm{d}(a^{-1}\mathrm{d}a + a^{-1}\omega a) + (a^{-1}\mathrm{d}a + a^{-1}\omega a) \wedge (a^{-1}\mathrm{d}a + a^{-1}\omega a) \\ =&\mathrm{d}a^{-1} \wedge \mathrm{d}a + \mathrm{d}a^{-1} \wedge \omega a + a^{-1}\mathrm{d}\omega\, a - a^{-1}\omega \wedge \mathrm{d}a \\ &+ a^{-1}\mathrm{d}a \wedge a^{-1}\mathrm{d}a + a^{-1}\mathrm{d}a \wedge a^{-1}\omega a + a^{-1}\omega \wedge \mathrm{d}a + a^{-1}\omega \wedge \omega a,\end{aligned}$$

since

$$a^{-1}a = I \implies \mathrm{d}a^{-1}a + a^{-1}\mathrm{d}a = 0 \implies \mathrm{d}a^{-1} = -a^{-1}\mathrm{d}a\, a^{-1},$$

we have

$$\begin{aligned}\tilde{\Omega} =& -a^{-1}\mathrm{d}a\, a^{-1} \wedge \mathrm{d}a - a^{-1}\mathrm{d}a\, a^{-1} \wedge \omega a + a^{-1}\mathrm{d}\omega\, a - a^{-1}\omega \wedge \mathrm{d}a \\ &+ a^{-1}\mathrm{d}a \wedge a^{-1}\mathrm{d}a + a^{-1}\mathrm{d}a \wedge a^{-1}\omega a + a^{-1}\omega \wedge \mathrm{d}a + a^{-1}\omega \wedge \omega a \\ =&a^{-1}(\mathrm{d}\omega + \omega \wedge \omega)a = a^{-1}\Omega a.\end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \tag{2.3}$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

2.2 General Affine Connections

Let $\pi : E \to M$ be a rank m vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

2.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 2.8. An (affine) connection ∇ on a vector bundle E is a map

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, E) &\to C^\infty(M, E) \\ (X, s) &\mapsto \nabla_X s \end{aligned}$$

satisfying

(C1) $$\nabla_{f_1Y_1+f_2Y_2}X = f_1\nabla_{Y_1}X + f_2\nabla_{Y_2}X,$$

(C2) $$\nabla_Y(\lambda_1X_1 + \lambda_2X_2) = \lambda_1\nabla_YX_1 + \lambda_2\nabla_YX_2,$$

(C3) $$\nabla_YfX = \mathrm{d}f(Y)\cdot X + f\nabla_YX.$$

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned} \nabla : \Omega^*(M, E) &\to \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla\eta, \end{aligned}$$

satisfying the Leibniz rule: for any $\alpha \in \Omega^q(M), \eta \in \Omega^*(M, E)$,

$$\nabla(\alpha \wedge \eta) = \mathrm{d}\alpha \wedge \eta + (-1)^q\alpha \wedge \nabla\eta.$$

Remark. The reason why we call ∇ an **affine** connection is that given two connections ∇_1 and ∇_2 on E, $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Remark. The existence of connections on E can be proved by using partitions of unity, which we left as an exercise.

Proposition 2.9. *Let ∇_1 and ∇_2 be two connections on E, then $\nabla_1 - \nabla_2$ is $C^\infty(M)$-linear in s, thus defines a section $A \in \Omega^1(M, \mathrm{End}(E))$. Conversely, for any $A \in \Omega^1(M, \mathrm{End}(E))$ and a connection ∇ on E, $\nabla + A$ is also a connection on E. Consequently, the space of connections on E is an affine space modeled on $\Omega^1(M, \mathrm{End}(E))$.*

Proof. For $s \in C^\infty(M, E), f \in C^\infty(M)$,

$$(\nabla_1 - \nabla_2)(fs) = \mathrm{d}f \otimes s + f\nabla_1 s - \mathrm{d}f \otimes s - f\nabla_2 s = f(\nabla_1 - \nabla_2)(s),$$

thus $\nabla_1 - \nabla_2$ is $C^\infty(M)$-linear in s. Conversely, for $A \in \Omega^1(M, \mathrm{End}(E))$ and a connection ∇ on E,

$$(\nabla + A)(fs) = \nabla(fs) + A(fs) = \mathrm{d}f \otimes s + f\nabla s + fAs = \mathrm{d}f \otimes s + f(\nabla + A)(s). \qquad \square$$

Curvature of connections

Proposition 2.10. $\nabla^2 : \Omega^*(M,E) \to \Omega^{*+2}(M,E)$ *is an* $\Omega^*(M)$*-linear map, i.e., for any* $\alpha \in \Omega^*(M), \eta \in \Omega^*(M,E)$,

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2\eta.$$

Proof. For $\alpha \in \Omega^q(M), \eta \in \Omega^*(M,E)$,

$$\begin{aligned}\nabla^2\big(\alpha \wedge \eta\big) =& \nabla(\mathrm{d}\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla\eta\big) \\ =& \mathrm{d}^2\alpha \wedge \eta + (-1)^{q+1}\mathrm{d}\alpha \wedge \nabla\eta + (-1)^q \mathrm{d}\alpha \wedge \nabla\eta + \alpha \wedge \nabla^2\eta \\ =& \alpha \wedge \nabla^2\eta.\end{aligned}$$

□

Definition 2.11. The curvature of ∇ is defined as $R := \nabla^2 \in \Omega^2(M, \mathrm{End}(E))$.

Metrics and metric compatibility

Definition 2.12. Let E be a (complex) vector bundle over M. A **(Hermitian) metric** on E is a smooth section g of $E^* \otimes E^*$ such that for any $x \in M$, g_x is a (Hermitian) inner product $\langle \cdot, \cdot \rangle_{E_x}$ on the fibre E_x.

Definition 2.13. A connection ∇ on E is called **compatible with the metric** g if for any $X \in C^\infty(M, TM)$ and $s_1, s_2 \in C^\infty(M, E)$,

$$X\langle s_1, s_2\rangle = \langle \nabla_X s_1, s_2\rangle + \langle s_1, \nabla_X s_2\rangle.$$

Remark. For a (complex) vector bundle E over M, we can always construct a (Hermitian) metric on E by using partitions of unity.

Remark. Given a (Hermitian) metric g on E, we can always construct a connection ∇ compatible with g by using partitions of unity. Note that the connection compatible with g is not unique.

Proposition 2.14. *Let* ∇_1 *and* ∇_2 *be two connections on* E *compatible with metric* g*, then* $\nabla_1 - \nabla_2$ *is skew-symmetric w.r.t.* g*, i.e.,* $\nabla_1 - \nabla_2 \in \Omega^1(M, \mathfrak{so}(E,g))$*. Conversely, for any* $A \in \Omega^1(M, \mathfrak{so}(E,g))$ *and a connection* ∇ *compatible with* g*,* $\nabla + A$ *is also compatible with* g*. Consequently, the space of connections on* E *compatible with* g *is an affine space modeled on* $\Omega^1(M, \mathfrak{so}(E,g))$*.*

Proof. For $s_1, s_2 \in C^\infty(M,E), X \in C^\infty(M,TM)$,

$$\begin{cases} X\langle s_1, s_2\rangle = \langle \nabla_{1,X} s_1, s_2\rangle + \langle s_1, \nabla_{1,X} s_2\rangle \\ X\langle s_1, s_2\rangle = \langle \nabla_{2,X} s_1, s_2\rangle + \langle s_1, \nabla_{2,X} s_2\rangle \end{cases}$$
$$\implies \langle (\nabla_1 - \nabla_2)_X s_1, s_2\rangle + \langle s_1, (\nabla_1 - \nabla_2)_X s_2\rangle = 0.$$

Conversely, for $A \in \Omega^1(M, \mathfrak{so}(E,g))$ and a connection ∇ compatible with g,

$$\begin{aligned}&\langle (\nabla + A)_X s_1, s_2\rangle + \langle s_1, (\nabla + A)_X s_2\rangle \\ =& \langle \nabla_X s_1, s_2\rangle + \langle s_1, \nabla_X s_2\rangle + \langle A(X)s_1, s_2\rangle + \langle s_1, A(X)s_2\rangle \\ =& X\langle s_1, s_2\rangle.\end{aligned}$$

□

2.2.2 Connection 1-form

Let E be a rank m vector bundle over M, and $\{e_i\}$ be a local frame of E on an open set $U \subset M$. Define ω_j^k by

$$\nabla_X e_j = \omega_j^k(X) e_k,$$

this 1-form matrix $\omega = (\omega_j^k)$ is called **the connection 1-form** of ∇ w.r.t. $\{e_i\}$. Then we can write the above equation in a more compact way:

$$\nabla(e_1, \ldots, e_m) = (e_1, \ldots, e_m)\,\omega = (e_1, \ldots, e_m) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^m \\ \vdots & & \vdots \\ \omega_m^1 & \cdots & \omega_m^m \end{pmatrix}.$$

Thus for any section $s = s^i e_i$, we have

$$\begin{aligned} \nabla s =& \nabla \left((e_1, \ldots, e_m) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right) = \Big(\nabla (e_1, \ldots, e_m) \Big) \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + (e_1, \ldots, e_m) \begin{pmatrix} \mathrm{d}s^1 \\ \vdots \\ \mathrm{d}s^m \end{pmatrix} \\ =& (e_1, \ldots, e_m) \left(\mathrm{d} \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} + \omega \begin{pmatrix} s^1 \\ \vdots \\ s^m \end{pmatrix} \right). \end{aligned}$$

So under a local frame, ∇ can be expressed as

$$\nabla = \mathrm{d} + \omega.$$

Proposition 2.15. *Suppose $\{\tilde{e}_i\}$ are another local frame of E on U, and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_i\}$ is $a = (a_i^j)$, i.e.*

$$(\tilde{e}_1, \ldots, \tilde{e}_m) = (e_1, \ldots, e_m)\, a,$$

Let $\tilde{\omega}$ be the connection 1-form w.r.t. $\{\tilde{e}_i\}$, then the connection 1-forms transform as

$$\tilde{\omega} = a^{-1}\mathrm{d}a + a^{-1}\omega a.$$

A local frame of E is the same as a local trivialization. Let $\{(U_\alpha, \psi_\alpha)\}$ be a local trivialization of E, $g_{\alpha\beta}$ be the transition functions. A section $s \in C^\infty(M, E)$ possesses a local representative s_α, which is an m-tuple of smooth functions on U_α. Also, ∇s possesses a local representative $(\nabla s)_\alpha$, which is an m-tuple of 1-forms on U_α. Just as in the preceding paragraph, we can write

$$(\nabla s)_\alpha = \mathrm{d}s_\alpha + \omega_\alpha s_\alpha, \tag{2.4}$$

where ω_α is an $m \times m$ matrix of 1-forms on U_α.

Since the connection is globally defined, the local representatives of ∇s must satisfy

$$\mathrm{d}s_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1}(\mathrm{d}s_\alpha + \omega_\alpha s_\alpha) \qquad \text{on} \ \ U_\alpha \cap U_\beta.$$

Apply $s_\alpha = g_{\alpha\beta} s_\beta$ to the above equation,

$$\mathrm{d}s_\beta + \omega_\beta s_\beta = g_{\alpha\beta}^{-1}\big(\mathrm{d}(g_{\alpha\beta} s_\beta) + \omega_\alpha g_{\alpha\beta} s_\beta\big)$$

$$
\begin{aligned}
&= g_{\alpha\beta}^{-1}\big((\mathrm{d}g_{\alpha\beta})s_\beta + g_{\alpha\beta}\mathrm{d}s_\beta + \omega_\alpha g_{\alpha\beta}s_\beta\big) \\
&= \mathrm{d}s_\beta + \Big(g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}\Big)s_\beta,
\end{aligned}
$$

Thus

$$
\omega_\beta = g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta}. \tag{2.5}
$$

The above discussion can be summarized as the following theorem:

Theorem 2.16. *Let $\{(U_\alpha,\psi_\alpha)\}$ be a local trivialization of E, with transition functions $g_{\alpha\beta}$, and ω_α be an $m\times m$ matrix of 1-forms on U_α satisfying the transformation formula (2.5). This information uniquely determines a connection ∇ on E.*

Let (E,g) be a real vector bundle equipped with a metric, we can choose the local frame $\{e_i\}$ on U to be orthonormal, then for any $X\in C^\infty(U,TU)$,

$$
0 = X\langle e_i,e_j\rangle = \langle \omega_i^k(X)e_k, e_j\rangle + \langle e_i, \omega_j^k(X)e_k\rangle = \omega_j^i(X) + \omega_i^j(X),
$$

so the 1-form matrix ω is skew-symmetric, i.e., $\omega\in\Omega^1(U,\mathfrak{so}(m))$.

Let $\{(U_\alpha,\psi_\alpha)\}$ be a local trivialization of (E,g). We can assume

$$
\psi_\alpha : \pi^{-1}(U_\alpha)\to U_\alpha\times\mathbb{R}^m
$$

is isometric on each fibre, where the metric on $U_\alpha\times\mathbb{R}^m$ is Euclidean metric on each fibre. Then the transition functions $g_{\alpha\beta}$ take values in $O(m)$, and each ω_α is in $\Omega^1(U_\alpha,\mathfrak{so}(m))$.

If E is a complex vector bundle and g is a Hermitian metric. Then all the discussion above still holds, except that ω_α is in $\Omega^1(U_\alpha,\mathfrak{u}(m))$ and $g_{\alpha\beta}$ takes values in $U(m)$.

Example 2.17. Let $\pi : L\to M$ be a complex line bundle, g be a Hermition metric on L and ∇ be a connection on L compatible with g. Choose a local trivialization $\{(U_\alpha,\psi_\alpha)\}$ of L such that $\psi_\alpha : \pi^{-1}(U_\alpha)\to U_\alpha\times\mathbb{C}$ is an isometry. Then $g_{\alpha\beta}$ takes values in $U(1)$, so we can write $g_{\alpha\beta}=e^{i\theta_{\alpha\beta}}$ for some real-valued function $\theta_{\alpha\beta}$ on $U_\alpha\cap U_\beta$. Meanwhile, ω_α takes values in $\mathfrak{u}(1)$, so we can write $\omega_\alpha = iA_\alpha$ for some real-valued 1-form A_α on U_α. For any section s of L, on U_α, we have

$$
(\nabla s)_\alpha = \mathrm{d}s_\alpha + \omega_\alpha s_\alpha, \qquad \omega_\alpha = iA_\alpha,
$$

On $U_\alpha\cap U_\beta$,

$$
\omega_\beta = g_{\alpha\beta}^{-1}\mathrm{d}g_{\alpha\beta} + g_{\alpha\beta}^{-1}\omega_\alpha g_{\alpha\beta} = e^{-i\theta_{\alpha\beta}}\mathrm{d}e^{i\theta_{\alpha\beta}} + e^{-i\theta_{\alpha\beta}}\omega_\alpha e^{i\theta_{\alpha\beta}} = \omega_\alpha + i\mathrm{d}\theta_{\alpha\beta}\Big(=\omega_\alpha - i\mathrm{d}\theta_{\beta\alpha}\Big)
$$

Recall $U(1)$-gauge theory in physics, the following table shows the correspondence between concepts in physics and differential geometry:

Physics	**Differential Geometry**
wave function ψ	section s of complex line bundle L
$\psi\mapsto\psi' = e^{i\chi}\psi$	$s_\beta = g_{\beta\alpha}s_\alpha$
covariant derivative	connection ∇ on L
$D'\psi' = e^{i\chi}D\psi$	$(\nabla s)_\beta = g_{\beta\alpha}(\nabla s)_\alpha$ [1]
$D = \mathrm{d} + iA$	$(\nabla s)_\alpha = \mathrm{d}s_\alpha + \omega_\alpha s_\alpha$
$A' = A - \mathrm{d}\chi$	$\omega_\beta = \omega_\alpha - i\mathrm{d}\theta_{\beta\alpha}$

[1] That is to say, ∇s is globally defined.

2.2.3 Horizontal Distribution

In this subsection, we give another interpretation of a connection on a vector bundle, namely a horizontal distribution, and prove that it is equivalent to the definition given in terms of covariant derivatives.

Let $\pi : E \to M$ be a rank m vector bundle, and $p \in E_x$ be a point in the fibre over $x \in M$. The propjection π induces a linear map $\pi_{*p} : T_pE \to T_xM$. We define $\mathcal{V}_p := \ker \pi_{*p}$, which is called the **vertical tangent space** at p. Thus we have a short exact sequence

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_pE \xrightarrow{\pi_*} T_{\pi(p)}M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_pE - \dim T_{\pi(p)}M = m.$$

The kernel of π_* form a subbundle of TE. We call it the **vertical subbundle** of TE and denote it by $\mathcal{V}$.

Proposition 2.18. *For each $p \in E$, there is an isomorphism $E_{\pi(p)} \cong \mathcal{V}_p$, which induces an isomorphism $\pi^*E \cong \mathcal{V}$.*

Proof. A point $(p, p') \in \pi^*E$ must satisfy $\pi(p) = \pi(p')$, so we can define a curve $\gamma(t) = p + tp'$ in E. Then $\gamma'(0) \in T_pE$ gives a tangent vector at p, this gives an injective morphism

$$\pi^*E \hookrightarrow TE, \quad (p, p') \mapsto (p, \gamma'(0)).$$

So we can view π^*E as a subbundle of TE. Moreover, since $\gamma(t) \subset E_{\pi(p)}$, we have $\pi(\gamma(t)) = \pi(p)$, thus $\pi_{*p}(\gamma'(0)) = 0$. Therefore, $\pi^*E \subset \ker \pi_*$

Since π_{*p} is a surjective linear map, we have

$$\dim \ker \pi_{*p} = \dim T_pE - \dim T_{\pi(p)}M = m = (\pi^*E)_p.$$

Thus $\pi^*E = \ker \pi_*$. □

For any $p \in E$, we can define a map

$$\iota_p : E_{\pi(p)} \to \mathcal{V}_p \subset T_pE, \quad p' \mapsto \gamma'(0),$$

where $\gamma(t) = p + tp'$ is a curve in E. Then ι_p is an isomorphism between $E_{\pi(p)}$ and $\mathcal{V}_p$.

Let π^*TM be the pull-back of TM by π, then $\pi_* : TE \to TM$ can be viewed as a morphism $\tilde{\pi}_* : TE \to \pi^*TM$. The above proposition implies the following short exact sequence of vector bundles over E:

$$0 \longrightarrow \mathcal{V} \hookrightarrow TE \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

In smooth category, all short exact sequences of vector bundles split, so we can find a subbundle $\mathcal{H} \hookrightarrow TE$ such that

$$TE = \mathcal{H} \oplus \mathcal{V}, \qquad \tilde{\pi}_* : \mathcal{H} \to \pi^*TM \text{ is an isomorphism.}$$

On the vector bundle E, we have two natural operations: scalar multiplication and addition.

- The scalar multiplication:

$$m : \mathbb{R} \times E \to E, \quad m(\lambda, p) = \lambda p.$$

When fixing λ, we get a map $m_\lambda : E \to E, m_\lambda(p) = \lambda p$, which is the scalar multiplication by λ.

- The addition:

$$a : E \times_M E \to E, \quad a(p, q) = p + q,$$

where $E \times_M E = \{(p, q) \in E \times E : \pi(p) = \pi(q)\}$.

Definition 2.19. A **horizontal distribution** of E is a smooth subbundle $\mathcal{H} \subset TE$ satisfying the following properties

1. There exists a splitting map $k : \pi^* TM \to TE$ such that $\tilde{\pi}_* \circ k = \mathrm{id}$.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook] & TE \arrow[r, shift left, "\tilde{\pi}_*"] & \pi^* TM \arrow[l, shift left, "k"] \arrow[r] & 0 \end{tikzcd}$$

That is to say,

(H1) $$TE = \mathcal{H} \oplus \mathcal{V}.$$

2. $\mathcal{H}$ is invariant under the scalar multiplication on E, i.e. for any $\lambda \in \mathbb{R}$, $p \in E$,

(H2) $$m_{\lambda * p} \mathcal{H}_p = \mathcal{H}_{\lambda p}.$$

3. $\mathcal{H}$ is invariant under the addition on E, i.e. for any $(p, q) \in E \times_M E$,

(H3) $$a_{*(p,q)}(\mathcal{H}_p, \mathcal{H}_q) = \mathcal{H}_{p+q}.$$

The last two conditions are saying that $\mathcal{H}$ is compatible with the linear structure of E.

There is no canonical way to choose a horizontal distribution, so a horizontal distribution is an extra structure on E.

In what follows, we reveal that covariant derivatives and horizontal distributions can actually determine each other.

Covariant Derivative determines Horizontal Distribution

Let ∇ be a connection on E. Given any smmoth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = x$, we say $\tilde{\gamma} : (-\varepsilon, \varepsilon) \to E$ is a **lift** of γ if $\pi(\tilde{\gamma}(t)) = \gamma(t)$ for all t.

Definition 2.20. A lift $\tilde{\gamma}$ of γ is called a **horizontal lift** if

$$\nabla_{\gamma'(t)} \tilde{\gamma}(t) = 0, \quad \forall t \in (-\varepsilon, \varepsilon).$$

Example 2.21. In the case of a Riemannian manifold (M, g), the horizontal lift is exactly the parallel transport.

Proposition 2.22. *For any (picewise) smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ and any point $p \in E_{\gamma(0)}$, there exists a unique horizontal lift $\tilde{\gamma}$ of γ starting at p.*

Sketch of proof. The proof is exactly the same as the existence and uniqueness of geodesics. The outline of the proof is to write the equations as a system of ODEs in a local frame; the conclusion then follows from the existence and uniqueness theory of linear ODEs. □

For any vector $u \in T_x M$ and any point $p \in E_x$, we can find a curve γ such that $\gamma(0) = x$ and $\gamma'(0) = u$, the horizontal lift $\tilde{\gamma}$ gives a tangent vector $\tilde{\gamma}'(0) \in T_p E$.

Fact 2.1. $\tilde{\gamma}'(0) \in T_p E$ is independent of the choice of γ.

Sketch of proof. By solving the ODEs, we know that $\tilde{\gamma}'(0)$ depends only on the derivative of γ at 0, which is $\gamma'(0) = u$. □

So there is a well-defined map

$$T_{\pi(p)} M \to T_p E, \quad (u^h)_p := \tilde{\gamma}'(0).$$

We call $(u^h)_p$ the **horizontal lift** of u at p. By definition,

$$\pi(\tilde{\gamma}(t)) = \gamma(t) \implies \pi_{*p}(u^h)_p = \pi_{*p}(\tilde{\gamma}'(0)) = \gamma'(0) = u.$$

So we obtain a linear injection from $T_{\pi(p)} M$ to $T_p E$, whose image forms a distribution on E denoted by $\mathcal{H}$, i.e.,

$$\mathcal{H} := \left\{ (p, (u^h)_p) \mid p \in E, u \in T_{\pi(p)} M \right\}.$$

Fact 2.2. $\mathcal{H}$ is a smooth subbundle of TE.

Sketch of proof. This fact relies on the smooth dependence of solutions of linear ODEs on initial conditions and parameters. □

Fact 2.3. $\mathcal{H}$ is invariant under the scalar multiplication and addition on E.

Proof. Let $p, q \in E$ such that $\pi(p) = \pi(q)$, $\lambda \in \mathbb{R}$, and γ be a curve such that $\gamma(0) = \pi(p)$. Let $\tilde{\gamma}_p, \tilde{\gamma}_q$ be the horizontal lift of γ starting at p, q respectively, then by the uniqueness of horizontal lift, $\lambda \tilde{\gamma}_p$ is the horizontal lift of γ starting at λp, and $\tilde{\gamma}_p + \tilde{\gamma}_q$ is the horizontal lift of γ starting at $p + q$. Thus

$$\begin{aligned} m_{\lambda * p} \tilde{\gamma}'(0) &= (\lambda \tilde{\gamma}_p)'(0) \in \mathcal{H}_{\lambda p}, \\ a_{*(p,q)}(\tilde{\gamma}_p'(0), \tilde{\gamma}_q'(0)) &= (\tilde{\gamma}_p + \tilde{\gamma}_q)'(0) \in \mathcal{H}_{p+q}. \end{aligned}$$ □

The above discussion is summarized in the following theorem.

Theorem 2.23. *$\mathcal{H}$ is a horizontal distribution of E.*

Horizontal Distribution determines Covariant Derivative

Conversaly, given a horizontal distribution $\mathcal{H}$, we can derive a connection ∇ on E as follows.

Definition 2.24. For any $p \in E$ and $u \in TE$, we can write $u = h(u) + v(u)$ where $h(u) \in \mathcal{H}$ and $v(u) \in \mathcal{V}$. We call $h(u)$ and $v(u)$ the horizontal and vertical component of u respectively.

For any $X \in C^\infty(M, TM)$ and $s \in C^\infty(M, E)$, the tangent map of s gives a map $s_* : TM \to TE$, then at any point $x \in M$ we can define $(\nabla_X s)(x)$ by

$$(\nabla_X s)(x) := \iota_{s(x)}^{-1}\big(v(s_{*x} X_x)\big).$$

Note that $\iota_{s(x)} : E_x \to \mathcal{V}_{s(x)}$ is the isomorphism defined above. The idea is taking the vertical component of $s_{*x}(X_x)$.

Theorem 2.25. ∇ *defined above is a connection on* E*.*

To prove this theorem, we need the following lemma.

Lemma 2.26. *For* $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ *and* $u, w \in E_{\pi(p)}$*, we have*

$$m_{\lambda * p} \iota_p(u) = \iota_{\lambda p}(\lambda u),$$
$$a_{*(p,q)}(\iota_p(u), \iota_q(w)) = \iota_{p+q}(u + w).$$

Proof. By the definition of ι_p,

$$m_{\lambda * p} \iota_p(u) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} m_\lambda(p + tu) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\lambda p + t\lambda u) = \iota_{\lambda p}(\lambda u),$$
$$a_{*(p,q)}(\iota_p(u), \iota_q(w)) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (p + tu) + (q + tw) = \iota_{p+q}(u + w).$$

□

Lemma 2.27. *For any* $\lambda \in \mathbb{R}$, $(p, q) \in E \times_M E$ *and* $u \in TE_p$, $w \in TE_q$*,*

$$h(m_{\lambda * p} u) = m_{\lambda * p}(h(u)), \quad v(m_{\lambda * p} u) = m_{\lambda * p}(v(u)),$$
$$h(a_{*(p,q)}(u, w)) = a_{*(p,q)}(h(u), h(w)), \quad v(a_{*(p,q)}(u, w)) = a_{*(p,q)}(v(u), v(w)).$$

Proof. Decompose u and $m_{\lambda * p} u$ into their horizontal and vertical components, we get

$$m_{\lambda * p}(h(u)) + m_{\lambda * p}(v(u)) = m_{\lambda * p}(u) = h(m_{\lambda * p} u) + v(m_{\lambda * p} u).$$

Since $\mathcal{H}$ is compatible with the linear structure of the vector bundle, we have $m_{\lambda * p}(h(u)) \in \mathcal{H}_{\lambda p}$. Besides, Lemma 2.26 implies $m_{\lambda * p}(v(u)) \in \mathcal{V}_{\lambda p}$, thus

$$m_{\lambda * p}(h(u)) = h(m_{\lambda * p} u), \qquad m_{\lambda * p}(v(u)) = v(m_{\lambda * p} u).$$

The proof for the addition is exactly the same. □

Lemma 2.28. *For* $\nu \in T_\lambda \mathbb{R} \cong \mathbb{R}$ *and* $u \in T_p E$*, we have*

$$m_{*(\lambda,p)}(\nu, u) = \nu \cdot \iota_p(p) + m_{\lambda * p} u.$$

Proof. let $\gamma(t)$ be a curve in E such that $\gamma(0) = p$ and $\gamma'(0) = u$, then

$$\begin{aligned} m_{*(\lambda,p)}(\nu, u) &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\lambda + t\nu)\gamma(t) \\ &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\lambda p + t\nu p) + \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \lambda\gamma(t) \\ &=\nu \cdot \iota_p(p) + m_{\lambda * p} u. \end{aligned}$$

□

Proof of Theorem 2.25. We need to verify that ∇ satisfies the properties of a connection.

- $\nabla_{f_1 X_1 + f_2 X_2} s = f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s$:

$$\begin{aligned} \nabla_{f_1X_1+f_2X_2} s(x) &= \iota_{s(x)}^{-1}\Big(v\big(s_{*x}(f_1X_1 + f_2X_2)\big)\Big) \\ &=\iota_{s(x)}^{-1}\Big(v\big(s_{*x}(f_1(x)X_{1x})\big) + v\big(s_{*x}(f_2(x)X_{2x})\big)\Big) \\ &=\iota_{s(x)}^{-1}\Big(f_1(x)v\big(s_{*x}X_{1x}\big) + f_2(x)v\big(s_{*x}X_{2x}\big)\Big) \\ &=f_1\nabla_{X_1}s(x) + f_2\nabla_{X_2}s(x). \end{aligned}$$

- $\nabla_X(\lambda_1 s_1 + \lambda_2 s_2) = \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2$:

  For $\mathbb{R}$-linearity,

$$\begin{aligned} \nabla_X(\lambda s)(x) &= \iota_{(\lambda s)(x)}^{-1}\Big(v\big((\lambda s)_{*x}(X_x)\big)\Big) \\ &=\iota_{(\lambda s)(x)}^{-1}\Big(v\big(m_{\lambda * s(x)} s_{*x}(X_x)\big)\Big) \\ &=\iota_{(\lambda s)(x)}^{-1}\Big(m_{\lambda * s(x)} v\big(s_{*x}(X_x)\big)\Big) \quad \text{(By Lemma 2.27)} \\ &=\iota_{(\lambda s)(x)}^{-1}\Big(\iota_{(\lambda s)(x)}(\lambda\nabla_X s(x))\Big) \quad \text{(By Lemma 2.26)} \\ &=\lambda\nabla_X s(x), \end{aligned}$$

  For additivity,

$$\begin{aligned} \nabla_X(s_1 + s_2)(x) &= \iota_{(s_1+s_2)(x)}^{-1}\Big(v\big((s_1 + s_2)_{*x}X_x\big)\Big) \\ &=\iota_{(s_1+s_2)(x)}^{-1}\Big(v\big(a_{*(s_1(x),s_2(x))}(s_{1*x}X_x, s_{2*x}X_x)\big)\Big) \\ &=\iota_{(s_1+s_2)(x)}^{-1}\Big(a_{*(s_1(x),s_2(x))}\big(v(s_{1*x}X_x), v(s_{2*x}X_x)\big)\Big) \quad \text{(By Lemma 2.27)} \\ &=\iota_{(s_1+s_2)(x)}^{-1}\Big(\iota_{(s_1+s_2)(x)}(\nabla_X s_1(x) + \nabla_X s_2(x))\Big) \quad \text{(By Lemma 2.26)} \\ &=\nabla_X s_1(x) + \nabla_X s_2(x). \end{aligned}$$

- $\nabla_X(fs) = \mathrm{d}f(X) \cdot s + f\nabla_X s$:

$$\begin{aligned} \nabla_X(fs)(x) &= \iota_{(fs)(x)}^{-1}\Big(v\big((fs)_{*x}X_x\big)\Big) \\ &=\iota_{(fs)(x)}^{-1}\Big(v\big(m_{f(x)*s(x)}(f_{*x}X_x, s_{*x}X_x)\big)\Big) \end{aligned}$$

$$
\begin{aligned}
&=\iota_{(fs)(x)}^{-1}\Big(\mathrm{d}f(X)_x\cdot\iota_{(fs)(x)}(s(x))+v\big(m_{f(x)*s(x)}s_{*x}X_x\big)\Big) \quad \text{(By Lemma 2.28)}\\
&=\iota_{(fs)(x)}^{-1}\Big(\iota_{(fs)(x)}\big(\mathrm{d}f(X)_x\cdot s(x)+f(x)\nabla_X s(x)\big)\Big) \quad \text{(By Lemma 2.26)}\\
&=\mathrm{d}f(X)_x\cdot s(x)+f(x)\nabla_X s(x).
\end{aligned}
$$

□

2.2.4 Parallel Transport

Let ∇ be a connection on E. For any (piecewise) smooth curve $\gamma:[a,b]\to M$ and any point $v\in E_{\gamma(a)}$, There exists a unique horizontal lift $\tilde{\gamma}_v$ of γ starting at v. Varying $v\in E_{\gamma(a)}$, we get different point $w=\tilde{\gamma}_v(b)\in E_{\gamma(b)}$. This gives a map

$$
P^\gamma: E_{\gamma(a)}\to E_{\gamma(b)},\quad P^\gamma(v):=\tilde{\gamma}_v(b).
$$

Proposition 2.29. *P^γ is a linear isomorphism.*

Proof. For any $v,w\in E_{\gamma(a)}$ and $\lambda\in\mathbb{R}$, the uniqueness of horizontal lift implies that

$$
\lambda\tilde{\gamma}_v=\tilde{\gamma}_{\lambda v},\qquad \tilde{\gamma}_v+\tilde{\gamma}_w=\tilde{\gamma}_{v+w}.
$$

Thus

$$
\begin{aligned}
P^\gamma(\lambda v)&=\tilde{\gamma}_{\lambda v}(b)=(\lambda\tilde{\gamma}_v)(b)=\lambda P^\gamma(v),\\
P^\gamma(v+w)&=\tilde{\gamma}_{v+w}(b)=(\tilde{\gamma}_v+\tilde{\gamma}_w)(b)=P^\gamma(v)+P^\gamma(w).
\end{aligned}
$$

So P^γ is a linear map. Let $\bar{\gamma}(t):=\gamma(a+b-t)$ be the reversed curve of γ, then one can check that $P^{\bar{\gamma}}:E_{\gamma(b)}\to E_{\gamma(a)}$ is the inverse map of P^γ. □

Definition 2.30. P^γ is called **the parallel transport** along γ.

Remark. If E possesses a metric g and ∇ is compatible with g, then $P^\gamma:E_{\gamma(a)}\to E_{\gamma(b)}$ is an isometry. The converse is also true: if P^γ is an isometry for any curve γ, then ∇ is compatible with g.

Remark. P^γ highly depends on the choice of γ, not only on the endpoints $\gamma(a)$ and $\gamma(b)$. In general, if γ_1 and γ_2 are two curves with the same endpoints, P^{γ_1} and P^{γ_2} can be very different. In other words, if γ is a loop, $P^\gamma:E_{\gamma(a)}\to E_{\gamma(a)}$ may not be the identity map.

Given a connection ∇ on E, for any two points $x,y\in M$, we can link x and y by a curve γ, then ∇ gives a linear isomorphism $P^\gamma:E_x\to E_y$. In this sense, ∇ connects any two fibers of E. This is the reason why ∇ is called a **connection**.

The key point is that if any curve γ linking x and y gives an isomorphism $\Phi^\gamma:E_x\to E_y$ and Φ^γ satisfies some natural properties, then we can define a connection ∇ on E such that $P^\gamma=\Phi^\gamma$. So intuitively, a connection can be thought of as a family of isomorphisms between fibers at endpoints of curves.

Proposition 2.31. *Let ∇ be a connection on E, $X_x\in T_xM$ be a tangent vector at x and $s:M\to E$ be a section of E. Then there exists a curve $\gamma:(-\varepsilon,\varepsilon)\to M$ such that $\gamma(0)=x$ and $\gamma'(0)=X_x$. If we denote $P_t:E_x\to E_{\gamma(t)}$ as the parallel transport along $\gamma|_{[0,t]}$, then*

$$
\nabla_{X_x}s=\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}P_t^{-1}(s(\gamma(t))). \tag{2.6}
$$

Proof. Let $e_1, \dots, e_m$ be a base of E_x and set $e_i(t) := P_t(e_i)$, then $e_1(t), \dots, e_m(t)$ is a base of $E_{\gamma(t)}$. We can write

$$s(\gamma(t)) = \sum_{i=1}^{m} s^i(t) e_i(t),$$

Then

$$\nabla_X s = \nabla_{\gamma'(0)} s = \sum_{i=1}^{m} \frac{\mathrm{d}s^i}{\mathrm{d}t}(0) e_i.$$

On the other hand,

$$P_t^{-1}(s(\gamma(t))) = \sum_{i=1}^{m} s^i(t) e_i,$$

so

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} P_t^{-1}(s(\gamma(t))) = \sum_{i=1}^{m} \frac{\mathrm{d}s^i}{\mathrm{d}t}(0) e_i = \nabla_X s. \qquad \square$$

Proposition 2.32. *Let $P^\gamma : E_{\gamma(a)} \to E_{\gamma(b)}$ be the parallel transport along γ, then P^γ satisfies the following properties:*

- *If the end point of γ_1 is the starting point of γ_2, we can concatenate γ_1 and γ_2 to get $\gamma_1 * \gamma_2$, then*

  (P1) $$P^{\gamma_1 * \gamma_2} = P^{\gamma_2} \circ P^{\gamma_1}.$$

- *If $\bar\gamma$ is the reversed curve of γ, then*

  (P2) $$P^{\bar\gamma} = (P^\gamma)^{-1}.$$

- *Let $\mathcal{U}$ be an open subset in $\mathbb{R}^k$ and $\gamma : \mathcal{U} \times [0,1] \to M$ is a smooth map. For any $u \in \mathcal{U}$, we denote $\gamma_u(t) := \gamma(u,t)$, then $\{\gamma_u\}$ is a family of curves smoothly depending on u. Let $s : \mathcal{U} \to E$ be a smooth map such that $s(u) \in E_{\gamma_u(0)}$. Then the map $\mathcal{U} \to E$, $u \mapsto P^{\gamma_u}(s(u))$ is also smooth. For simplicity, we state this conclusion as*

  (P3) $$P^\gamma \text{ depends smoothly on } \gamma.$$

- *The last property is a bit technical. Let $\gamma_1, \gamma_2 : [0,1] \to M$ be two curves such that $\gamma_1(0) = \gamma_2(0)$, and $\gamma_1'(0) = \gamma_2'(0)$, Let $P_t^{\gamma_i}$ be the parallel transport along $\gamma_i|_{[0,t]}$, then*

  $$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} P_t^{\gamma_1} = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} P_t^{\gamma_2}.$$

  For simplicity, we state this conclusion as

  (P4) *If γ_1, γ_2 have contact of order* 1*, then $P^{\gamma_1}, P^{\gamma_2}$ have contact of order* 1.

Proof. (P1) and (P2) are straightforward from the uniqueness of solution of ODEs.

(P3) relies on the smooth dependence of solutions of ODEs on initial conditions and parameters.

For (P4), given any $v \in E_{\gamma_1(0)}$, $\tilde\gamma_{i,v}(t) := P_t^{\gamma_i}(v)$ is the horizontal lift of γ_i starting at v. The proposition is equivalent to saying that $\tilde\gamma_{1,v}'(0) = \tilde\gamma_{2,v}'(0)$, which can be verified by calculating under a local trivialization. $\square$

Parallel Transport determines Covariant Derivative

Theorem 2.33. *Let E be a vector bundle over M. Suppose there is a family of isomorphisms*

$$\left\{\Phi^\gamma : E_{\gamma(0)} \to E_{\gamma(1)} \,\middle|\, \gamma : [0,1] \to M\right\}$$

satisfying (P1)-(P4), then there exists a unique connection ∇ on E such that $P^\gamma = \Phi^\gamma$ for any curve γ.

Proof. For any vector field $X \in C^\infty(M, TM)$ and section $s \in C^\infty(M, E)$, at $x \in M$, we can find a curve $\gamma : (-\varepsilon, \varepsilon) \to M$ such that $\gamma(0) = x$ and $\gamma'(0) = X_x$. We define $(\nabla_X s)(x)$ by the formular (2.6), i.e.,

$$(\nabla_X s)(x) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\Phi_t^\gamma)^{-1}(s(\gamma(t))).$$

By (P4), this definition does not depend on the choice of γ. By (P3), $\nabla_X s$ is a smooth section of E.

We can verify that ∇ satisfies (C2) and (C3) by direct calculation. For (C1), we need to verify that $\nabla_{X_x} s$ is $\mathbb{R}$-linear in X_x. Let $\varphi(X_x) = \nabla_{X_x} s$, then φ is a map from $T_x M$ to E_x. For any $\lambda \in \mathbb{R}$, set $\gamma_\lambda(t) = \gamma(\lambda t)$, then $\gamma_\lambda(0) = x$ and $\gamma'_\lambda(0) = \lambda X_x$, so

$$\begin{aligned}
\varphi(\lambda X_x) =& \nabla_{\gamma'_\lambda(0)} s = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\Phi_t^{\gamma_\lambda})^{-1}(s(\gamma_\lambda(t))) \\
=& \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\Phi_{\lambda t}^{\gamma})^{-1}(s(\gamma(\lambda t))) \\
=& \lambda \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} (\Phi_t^{\gamma})^{-1}(s(\gamma(t))) \\
=& \lambda \nabla_{\gamma'(0)} s = \lambda \varphi(X_x).
\end{aligned}$$

It is easy to verify that φ is additive, thus φ is linear.

Finally, we can verify that $P^\gamma = \Phi^\gamma$ for any curve γ by direct calculation. □

Above theorem actually says that the covariant derivative ∇ is the infinitesimal version of the parallel transport P^γ.

Parallel Transport and Horizontal Distribution

Let $\pi : E \to M$ be a vector bundle. Given a family of parallel transports $\{P^\gamma\}$, we can define a horizontal distribution $\mathcal{H}$ by

$$\mathcal{H}_p := \left\{ \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} P_t^\gamma(p) \,\middle|\, \gamma : [0,1] \to M \text{ with } \gamma(0) = p \right\} \subset T_p E.$$

Varying $p \in E$, we get a distribution $\mathcal{H}$ of E. Note that this definition is consistent with the previous one. Thus we have

Theorem 2.34. *If $\{P^\gamma\}$ is a family of parallel transports satisfying (P1)-(P4), then $\mathcal{H}$ satisfies (H1)-(H3), thus $\mathcal{H}$ is a horizontal distribution of E.*

One can view $\mathcal{H}$ as the collection of all directions one can possibly move via parallel transports.

Conversely, given a horizontal distribution $\mathcal{H}$, we can define the horizontal lift, then the parallel transport. Expilicitly, for any curve $\gamma : [a, b] \to M$ and any point $v \in E_{\gamma(a)}$, there exists a unique curve $\tilde{\gamma}_v^{\mathcal{H}} : [a, b] \to E$ such that

$$\pi(\tilde{\gamma}_v^{\mathcal{H}}(t)) = \gamma(t), \quad \tilde{\gamma}_v^{\mathcal{H}}(a) = v, \quad \tilde{\gamma}_v^{\mathcal{H}\prime}(t) \in \mathcal{H}_{\tilde{\gamma}_v^{\mathcal{H}}(t)}.$$

The last condition can be translated into an ODE. Since we have a decomposition $TE = \mathcal{H} \oplus \mathcal{V}$, we can denote the projection to $\mathcal{V}$ by v. Then

$$\tilde{\gamma}_v^{\mathcal{H}\prime}(t) \in \mathcal{H}_{\tilde{\gamma}_v^{\mathcal{H}}(t)} \iff v(\tilde{\gamma}_v^{\mathcal{H}\prime}(t)) = 0,$$

which can be written as a linear ODEs under a local trivialization. Thus we can solve $\tilde{\gamma}_v^{\mathcal{H}}$ by the theory of ODEs. Now we can define $P^\gamma(v) := \tilde{\gamma}_v^{\mathcal{H}}(b)$.

Proposition 2.35. *Fix a curve $\gamma : [0, 1] \to M$ connecting x and y, for any $v, w \in E_x$, $\lambda \in \mathbb{R}$,*

$$\begin{aligned} \tilde{\gamma}_{v+w}^{\mathcal{H}}(t) &= \tilde{\gamma}_v^{\mathcal{H}}(t) + \tilde{\gamma}_w^{\mathcal{H}}(t), \\ \tilde{\gamma}_{\lambda v}^{\mathcal{H}}(t) &= \lambda \tilde{\gamma}_v^{\mathcal{H}}(t). \end{aligned}$$

Thus $P^\gamma : E_x \to E_{\gamma(1)}$ is a linear map. Moreover, P^γ is an isomorphism with inverse $P^{\bar{\gamma}}$ where $\bar{\gamma}$ is the reversed curve of γ.

Proof. The first two equalities are implied by the linearity of $\mathcal{H}$, and the last one can be verified by direct calculation. □

Theorem 2.36. *If $\mathcal{H}$ is a horizontal distribution of E, then the parallel transport defined above satisfies (P1)-(P4).*

Proof. The proof is similar to the previous one. □

In summary, three different concepts, i.e., covariant derivative ∇, parallel transport $\{P^\gamma\}$ and horizontal distribution $\mathcal{H}$, are equivalent to each other.

$$\text{covariant derivative} \iff \text{parallel transport} \iff \text{horizontal distribution.}$$

In fact, both ∇ and $\mathcal{H}$ can be viewed as the infinitesimal version of $\{P^\gamma\}$.

Exercises

Exercise 2.1. Given a connection ∇, we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s)-tensor field T, the covariant derivative $\nabla_Z T$ is

$$\begin{aligned}(\nabla_Z T)(X_1, \ldots, X_r, \omega^1, \ldots, \omega^s) : = &Z(T(X_1, \ldots, X_r, \omega^1, \ldots, \omega^s)) \\ &- \sum_{i=1}^{r} T(X_1, \ldots, \nabla_Z X_i, \ldots, X_r, \omega^1, \ldots, \omega^s) \\ &- \sum_{j=1}^{s} T(X_1, \ldots, X_r, \omega^1, \ldots, \nabla_Z \omega^j, \ldots, \omega^s).\end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X,Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X,Y]} T.$$

We define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s+1)$-tensor field

$$(\nabla_{\text{tensor}} T)(Z, \cdots, X_i, \cdots, \cdots, \omega^j, \cdots) := (\nabla_Z T)(\cdots, X_i, \cdots, \cdots, \omega^j, \cdots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

Prove that for a torsion-free connection, ∇_{tensor} satisfies **the Ricci identity**

$$(\nabla^2_{\text{tensor}} T)(X, Y, \cdots, \cdots) - (\nabla^2_{\text{tensor}} T)(Y, X, \cdots, \cdots) = (R(X,Y)T)(\cdots, \cdots)$$

Exercise 2.2. For a vector field $Z \in C^\infty(M, TM)$, suppose ∇ is torsion-free. Both $\nabla^2 Z$ and $\nabla^2_{\text{tensor}} Z$ are $(1, 2)$-tensor fields, but by Ricci identity, we have

$$(\nabla^2_{\text{tensor}} Z)(X, Y) - (\nabla^2_{\text{tensor}} Z)(Y, X) = R(X,Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?

Exercise 2.3 (Torsion 2-form)**.** Let ∇ be a connection on TM, $\{e_i\}$ be a local frame of TM, $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. Define

$$\tau^i := \mathrm{d}\theta^i + \omega^i_j \wedge \theta^j.$$

If we arrange $\theta = (\theta^1, \ldots, \theta^n)^T$ as a column vector, then the above formula can be written as

$$\tau = \mathrm{d}\theta + \omega \wedge \theta.$$

then τ is called the **torsion 2-form** of ∇ w.r.t. $\{e_i\}$.

Let $T(e_i, e_j) = T^k_{ij} e_k$, where T is the torsion of ∇. Prove that

$$\tau^k(e_i, e_j) = T^k_{ij}.$$

Thus

$$\tau^k = \frac{1}{2} T^k_{ij} \theta^i \wedge \theta^j.$$

Exercise 2.4 (Levi-Civita connection)**.** Let (M, g) be a Riemannian manifold, ∇ be a connection on TM. Show that

1. ∇ is torsion-free if and only if $\tau = 0$, i.e.,
$$\mathrm{d}\theta^i + \omega^i_j \wedge \theta^j = 0.$$
2. If $\{e_i\}$ is a local orthonormal frame, then ∇ is compatible with g if and only if ω is skew-symmetric, i.e.,
$$\omega^i_j + \omega^j_i = 0.$$
3. Under a local orthonormal frame, there exists a unique connection 1-form ω satisfying
$$\mathrm{d}\theta^i + \omega^i_j \wedge \theta^j = 0, \quad \omega^i_j + \omega^j_i = 0.$$
This is the existence and uniqueness of the Levi-Civita connection.

Exercise 2.5. Let ∇ be a connection on TM, $\{e_i\}$ be a local frame of TM, $\{\theta^i\}$ be the corresponding dual coframe and ω be the connection 1-form of ∇ w.r.t. $\{e_i\}$. The following two equations

$$\begin{aligned}\mathrm{d}\theta =& \tau - \omega \wedge \theta,\\ \mathrm{d}\omega =& \Omega - \omega \wedge \omega,\end{aligned}$$

are called the **structure equations**. Show that τ and Ω satisfy

$$\begin{aligned}\mathrm{d}\tau =& \Omega \wedge \theta - \omega \wedge \tau,\\ \mathrm{d}\Omega =& \Omega \wedge \omega - \omega \wedge \Omega.\end{aligned}$$

These two equations are called the **Bianchi identities**. Furthermore, show that

$$\mathrm{d}(\Omega^k) = \Omega^k \wedge \omega - \omega \wedge \Omega^k.$$

Exercise 2.6. Show that if ∇ is torsion-free, then $\mathrm{d}\tau = \Omega \wedge \theta - \omega \wedge \tau$ is equivalent to **the first Bianchi identity**

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0.$$

Exercise 2.7. Show that if ∇ is torsion-free, then $\mathrm{d}\Omega = \Omega \wedge \omega - \omega \wedge \Omega$ is equivalent to **the second Bianchi identity**

$$(\nabla_X R)(Y,Z) + (\nabla_Y R)(Z,X) + (\nabla_Z R)(X,Y) = 0.$$

Exercise 2.8. For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega^k_j = \Gamma^k_{ij}\mathrm{d}x^i$, show that the transformation formula

$$\tilde{\omega} = a^{-1}\mathrm{d}a + a^{-1}\omega a$$

reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}^k_{ij} = \frac{\partial \tilde{x}^k}{\partial x^l}\frac{\partial x^m}{\partial \tilde{x}^i}\frac{\partial x^n}{\partial \tilde{x}^j}\Gamma^l_{mn} + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j}\frac{\partial \tilde{x}^k}{\partial x^l}.$$

Exercise 2.9. Let ∇ be a connection on TM. It determines a horizontal distribution $\mathcal{H}$ on TM. Then $T(TM) = \mathcal{V} \oplus \mathcal{H}$, where $\mathcal{V}$ is the vertical bundle of TM. Show that ∇ is torsion-free if and only if for any $X, Y \in C^\infty(M, TM)$,

$$[X^h, Y^h] = [X,Y]^h.$$

3 Principal Bundles and Connections

3.1 Principal Bundles

Principal bundles are both a special type of fiber bundle and the most important one. In the following, we will introduce the definition of principal bundles and some basic structures on them.

Definition 3.1 (G-action)**.** A **smooth right action** of a Lie group G on a manifold P is a smooth map

$$\mu : M \times G \to M,$$

denoted by $\mu(x, g) := x \cdot g$, such that for all $x \in M$ and $g, h \in G$,

$$x \cdot e = x, \quad (x \cdot g) \cdot h = x \cdot (gh).$$

where e is the identity element of G. A manifold M with a smooth right G-action is called a (**right**) G**-manifold**.

We say the action is **free** if for all $x \in M$, $x \cdot g = x$ implies $g = e$.

Definition 3.2 (G-equivariant map)**.** Let M and N be two G-manifolds. A smooth map $f : M \to N$ is called G**-equivariant** if for all $x \in M$ and $g \in G$,

$$f(x \cdot g) = f(x) \cdot g.$$

Definition 3.3 (Principal G-bundle)**.** A **principal** G**-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a free right G-action on P such that for each $x \in M$, the local trivialization

$$\phi_U : \pi^{-1}(U) \to U \times G,$$

is G-equivariant, where the right G-action on $U \times G$ is given by

$$(x, h) \cdot g = (x, hg), \quad \forall x \in U, g, h \in G.$$

By a **local trivialization** we mean a open neighborhood U of x and a diffeomorphism $\phi_U : \pi^{-1}(U) \to U \times G$ satisfying the following commutative diagram:

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "\phi_U"] \arrow[dr, "\pi"'] & & U \times G \arrow[dl, "\mathrm{pr}_1"] \\
& U &
\end{tikzcd}$$

where $\mathrm{pr}_1 : U \times G \to U$ is the projection map.

Example 3.4. Let G be a Lie group and M be a smooth manifold, then the trivial bundle $p_1 : M \times G \to M$ is a principal G-bundle over M.

Example 3.5. If G is a finite group equipped with the discrete topology, then a principal G-bundle is exactly a covering space.

Example 3.6. Let G be a Lie group and H be a closed subgroup of G. Then the quotient space G/H is a smooth manifold and the projection map $\pi : G \to G/H$ is a principal H-bundle over G/H. For example, the Hopf fibration $S^1 \to S^3 \to S^2$.

Example 3.7. Let $\pi : E \to M$ be a rank m real vector bundle over M. Then the frame bundle $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}(m, \mathbb{R})$-bundle over M. If E is a complex vector bundle, then $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}(m, \mathbb{C})$-bundle over M.

Proposition 3.8. *Let $\pi : P \to M$ be a principal G-bundle. Then the right G-action on P is transitive on each fiber.*

Proof. Since G acts transitively on $\{x\} \times G$ and the fiber diffeomorphism $\phi_{U,x} : P_x \to G$ is G-equivariant, thus G acts transitively on P_x. □

Lemma 3.9. *For any group G, all right G-equivariant maps are exactly the left translations.*

Proof. Let $f : G \to G$ be a right G-equivariant map, then for all $g \in G$,

$$f(g) = f(e \cdot g) = f(e) \cdot g = l_{f(e)}(g),$$

where $l_{f(e)} : G \to G$ is the left translation by $f(e)$. □

Definition 3.10 (Cocycle definition)**.** A **principal G-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a local trivialization $\{(U_\alpha, \phi_\alpha)\}$ such that for each $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $g_{\alpha\beta} : U_{\alpha\beta} \to G$ such that

$$\phi_\alpha \circ \phi_\beta^{-1}(x, h) = (x, g_{\alpha\beta}(x)h), \quad \forall (x, h) \in U_{\alpha\beta} \times G.$$

In addition, the transition functions $\{g_{\alpha\beta}\}$ satisfy the following cocycle condition:

$$g_{\alpha\beta}(x)g_{\beta\gamma}(x) = g_{\alpha\gamma}(x), \quad \forall x \in U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset.$$

Remark. By Lemma 3.9, one can easily check that the two definitions of principal bundles are equivalent. These two definitions characterize G-principal bundles from different perspectives.

Fundamental Vector Fields

Suppose P is a G-manifold, then for each $A \in \mathfrak{g}$, we can define a vector field $\underline{A}$ on P by

$$\underline{A}_p := \frac{d}{dt}\bigg|_{t=0} p \cdot \exp(tA), \quad \forall p \in P.$$

The vector field $\underline{A}$ is called the **fundamental vector field** associated to A.

Fact 3.1. For each $A \in \mathfrak{g}$, $\underline{A}$ is a C^∞ vector field on P.

The fundamental vector field construction gives rise to a map

$$\sigma : \mathfrak{g} \to \mathfrak{X}(P),^1 \quad \sigma(A) := \underline{A}.$$

For $p \in P$, define $j_p : G \to P$ by $j_p(g) := p \cdot g$, then $j_{p*} := j_{p*e}$ is

$$j_{p*}(A) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} j_p(\exp(tA)) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} p \cdot \exp(tA) = \underline{A}_p.$$

This gives another discription of the fundamental vector field: $\underline{A}_p = j_{p*}(A)$. It also implies that the map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is $\mathbb{R}$-linear. In fact, we have

[1] $\mathfrak{X}(P) := C^\infty(P, TP)$ is the space of smooth vector fields on P.

Fact 3.2. The map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is a Lie algebra homomorphism, i.e.,

$$[\underline{A}, \underline{B}] = \underline{[A, B]}, \quad \forall A, B \in \mathfrak{g}.$$

Fact 3.3. Let P be a G-manifold, then the fundamental vector fields $\underline{A}$ vanishes at p if and only if $A \in \mathrm{Lie}(\mathrm{Stab}(p))$. Thus for any $p \in P$, the kernel of $j_{p*} : \mathfrak{g} \to T_pP$ is exactly $\mathrm{Lie}(\mathrm{Stab}(p))$. In particular, if the G-action on P is free, then j_{p*} is injective for all $p \in P$.

Example 3.11. Let G acts on itself by right translation, then the fundamental vector field $\underline{A}$ is exactly the left-invariant vector field generated by A.

For $g \in G$, let $c_g : G \to G$ be the conjugation map $c_g(h) := ghg^{-1}$. Then the differential of c_g at the identity element e is the adjoint representation $\mathrm{Ad}_g := (c_g)_* : \mathfrak{g} \to \mathfrak{g}$.

Proposition 3.12. *For each $A \in \mathfrak{g}$ and $g \in G$, we have*

$$(r_g)_*\underline{A} = \underline{\mathrm{Ad}_{g^{-1}}(A)}.$$

where $r_g : P \to P$ is the right translation by g, i.e., $r_g(p) := p \cdot g$.

Proof. For each $p \in P$ and $h \in G$, we have

$$(r_g \circ j_p)(h) = r_g(p \cdot h) = (p \cdot h) \cdot g = pg \cdot (g^{-1}hg) = (j_{pg} \circ c_{g^{-1}})(h).$$

By the chain rule,

$$(r_g)_*(\underline{A}_p) = (r_g)_* j_{p*}(A) = j_{pg*}(c_{g^{-1}})_*(A) = j_{pg*}(\mathrm{Ad}_{g^{-1}}(A)) = \underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}.$$ □

3.1.1 Vertical Subbundle of TP

Let $\pi : P \to M$ be a principal G-bundle, at each $p \in P$, the differential $\pi_{*p} : T_pP \to T_{\pi(p)}M$ is surjective. We define $\mathcal{V}_p := \ker \pi_{*p} \subset T_pP$ to be the **vertical tangent space** at p. Then $\mathcal{V} := \bigsqcup_{p \in P} \mathcal{V}_p$ is a subbundle of TP, called the **vertical subbundle** of TP. An element of $\mathcal{V}_p$ is called a **vertical tangent vector** at p.

We have a short exact sequence of vector spaces

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_pP \xrightarrow{\pi_{*p}} T_{\pi(p)}M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_pP - \dim T_{\pi(p)}M = \dim G.$$

Proposition 3.13. *For any $A \in \mathfrak{g}$, the fundamental vector field $\underline{A}$ is vertical at every point $p \in P$.*

Proof. Since $(\pi \circ j_p)(g) = \pi(p \cdot g) = \pi(p)$ is a constant map for all $g \in G$, we have $\pi_{*p}(\underline{A}_p) = \pi_{*p} j_{p*}(A) = (\pi \circ j_p)_*(A) = 0$. □

This proposition implies that $j_{p*}(\mathfrak{g}) \subset \mathcal{V}_p$ for all $p \in P$. In fact, we have the following stronger result.

Proposition 3.14. *For each $p \in P$, the map $j_{p*} : \mathfrak{g} \to \mathcal{V}_p$ is an isomorphism.*

Proof. Since G acts freely on P, the map j_{p*} is injective. On the other hand, $\dim \mathcal{V}_p = \dim G = \dim \mathfrak{g}$, thus j_{p*} is an isomorphism. □

Corollary 3.15. *The vertical vectors at $p \in P$ are exactly the fundamental vector at p. Moreover, the vertical subbundle $\mathcal{V}$ of TP is isomorphic to the trivial bundle $P \times \mathfrak{g}$.*

Proof. The first statement follows from the previous proposition. For the second statement, let $A_1, \ldots, A_l$ be a basis of $\mathfrak{g}$, where $l = \dim G$, then $\underline{A}_1, \ldots, \underline{A}_l$ is a global frame of $\mathcal{V}$, thus $\mathcal{V}$ is a trivial bundle. □

The map $\pi_* : TP \to TM$ induces a bundle map $\tilde{\pi}_* : TP \to \pi^*TM$ over P. Then we have a short exact sequence of vector bundles over P:

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

3.2 Connections on Principal Bundles

3.2.1 Horizontal Distributions

In the category of smooth vector bundles, all short exact sequence of vector bundles splits. Corresponding to the short exact sequence appearing in the previous chapter, there exists a smooth subbundle $\mathcal{H}$ of TP such that $TP = \mathcal{V} \oplus \mathcal{H}$. We call $\mathcal{H}$ a **horizontal distribution** of TP.

In general, there is no canonical choice of $\mathcal{H}$. A choice of $\mathcal{H}$ is the same as a splitting $k : \pi^*TM \to TP$ such that $\tilde{\pi}_* \circ k = \mathrm{id}_{\pi^*TM}$.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook] & TP \arrow[r, shift left, "\tilde{\pi}_*"] & \pi^*TM \arrow[l, shift left, "k"] \arrow[r] & 0 \end{tikzcd}$$

We say $\mathcal{H}$ is **right-invariant** if for all $g \in G$ and $p \in P$, $r_{g*p}(\mathcal{H}_p) = \mathcal{H}_{pg}$, where $r_g : P \to P$, $r_g(p) = p \cdot g$.

3.2.2 Ehresmann Connections

Let $\pi : P \to M$ be a principal G-bundle, $\mathcal{H}$ be a horizontal distribution of TP. For each $p \in P$, we define a projection map $v : T_pP = \mathcal{V}_p \oplus \mathcal{H}_p \to \mathcal{V}_p$. For $Y_p \in T_pP$, we call $v(Y_p)$ the **vertical component** of Y_p.[1]

Let $\iota : \mathcal{V} \hookrightarrow TP$ be the inclusion map, then $v \circ \iota = \mathrm{id}_{\mathcal{V}}$. We actually obtain another splitting of the short exact sequence:

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook, shift left, "\iota"] & TP \arrow[l, shift left, "v"] \arrow[r, "\tilde{\pi}_*"] & \pi^*TM \arrow[r] & 0 \end{tikzcd}$$

Recall that for each $p \in P$, the map $j_p : G \to P$ defined by $j_p(g) = p \cdot g$. We define a $\mathfrak{g}$-valued 1-form ω on P by

$$\omega_p := j_{p*}^{-1} \circ v : T_pP \xrightarrow{v} \mathcal{V}_p \xrightarrow{j_{p*}^{-1}} \mathfrak{g}.$$

[1]Although $\mathcal{V}_p$ is intrinsically defined, the vertical component of a tangent vector depends on the choice of $\mathcal{H}_p$.

In terms of ω, the vertical component of $Y_p \in T_pP$ is given by

$$v(Y_p) = j_{p*}(\omega_p(Y_p)).$$

In summary, given a projection $v : TP \to \mathcal{V}$, we induce the following composition of maps:

$$\omega : TP \xrightarrow{v} \mathcal{V} \xrightarrow{j_*^{-1}} P \times \mathfrak{g} \xrightarrow{\mathrm{pr}_2} \mathfrak{g} \quad \rightsquigarrow \quad \omega \in \Omega^1(P, \mathfrak{g}).$$

So the information of the horizontal distribution $\mathcal{H}$ is encoded in ω.

Theorem 3.16. *If $\mathcal{H}$ is a smooth, right-invariant horizontal distribution, then the $\mathfrak{g}$-valued 1-form ω defined above satisfies the following properties:*

1. $\omega_p(\underline{A}_p) = A$, $\forall\, p \in P$, $A \in \mathfrak{g}$.
2. *(G-equivalence)* $(r_g)^*\omega = (\mathrm{Ad}_{g^{-1}})\omega$, $\forall\, g \in G$.
3. *ω is smooth.*

We say a $\mathfrak{g}$-valued 1-form ω on P satisfying the above properties is an ***Ehresmann connection*** *or simply a* ***connection*** *on P.*

Proof. *1.* Since $\underline{A}$ is already vertical,

$$\omega_p(\underline{A}_p) = j_{p*}^{-1}(v(\underline{A}_p)) = j_{p*}^{-1}(\underline{A}_p) = A.$$

2. For each $p \in P$ and $Y_p \in T_pP$, we need to show

$$\omega_{pg}((r_g)_*Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

Since both side are linear in Y_p, we only need to check the above equality for $Y_p \in \mathcal{V}_p$ and $Y_p \in \mathcal{H}_p$ respectively.

If $Y_p \in \mathcal{V}_p$, then $Y_p = \underline{A}_p$ for some $A \in \mathfrak{g}$, thus

$$\omega_{pg}(r_{g*}\underline{A}_p) = \omega_{pg}(\underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}) = \mathrm{Ad}_{g^{-1}}(A) = (\mathrm{Ad}_{g^{-1}})\omega_p(\underline{A}_p).$$

If $Y_p \in \mathcal{H}_p$, by the right-invariance of $\mathcal{H}$, we have $r_{g*}(Y_p) \in \mathcal{H}_{pg}$, thus

$$\omega_{pg}(r_{g*}Y_p) = 0 = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

3. Omit. □

Viewing ω as a map $\omega : TP \to \mathfrak{g}$. Both TP and $\mathfrak{g}$ are G-manifolds, where

$$g \cdot Y_p := (r_g)_*Y_p, \quad g \cdot A := \mathrm{Ad}_{g^{-1}}(A).$$

Then the G-equivalence condition of ω

$$\omega_{pg}((r_g)_*Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p),$$

can be rephrased as $\omega : TP \to \mathfrak{g}$ is a G-equivariant map.

3.2.3 The Horizontal Distribution of an Ehresmann Connection

In the previous section we have shown that a smooth, right-invariant horizontal distribution $\mathcal{H}$ on P induces an Ehresmann connection ω. We now prove the converse.

Theorem 3.17. *If ω is an Ehresmann connection on the principal G-bundle $\pi : P \to M$, then $\mathcal{H}_p := \ker \omega_p$, $p \in P$ defines a smooth, right-invariant horizontal distribution on P.*

Proof. We need to verify three properties:
1. For each $p \in P$, there is an exact sequence

$$0 \longrightarrow \mathcal{H}_p \hookrightarrow T_pP \xrightarrow{\omega_p} \mathfrak{g} \longrightarrow 0$$

The map $j_{p*} : \mathfrak{g} \to \mathcal{V}_p \subset T_pP$ provides a splitting of the above sequence, thus

$$T_pP \cong \mathfrak{g} \oplus \mathcal{H}_p \cong \mathcal{V}_p \oplus \mathcal{H}_p.$$

2. Suppose $Y_p \in \mathcal{H}_p$, then $\omega_p(Y_p) = 0$. By the G-equivalence of ω, we have

$$\omega_{pg}(r_{g*}Y_p) = (r_g^*\omega)_p(Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p) = 0.$$

Hence $r_{g*}Y_p \in \mathcal{H}_{pg}$.
3. $\mathcal{H}$ is a smooth distribution. We omit the details here. $\square$